Finite-Length Block Separation in Cirac–Perez-Garcia–Schuch–Verstraete 2017 Proposition IV.3

2026-04-30

1 The assertion

The finite-length block-separation step in Proposition IV.3 of Cirac, Pérez-García, Schuch, and Verstraete is the point at issue [CPGSV16, CPGSV17].[1]

Let A be a tensor in canonical form with bond-dimension parameter D, and let $A_1, \ldots, A_g$ be a basis of normal tensors for A. The paper characterizes such a basis by minimality: no two chosen normal tensors may be related by an invertible similarity together with a phase. It then defines block-injective canonical form as the statement that, after passing to these blocks, simultaneous word evaluations span the whole product algebra

$$\bigoplus_{j=1}^{g} M_{D_j}(\mathbb{C}).$$

Equivalently, for every block tuple $X = (X_j)_j$ there should be a linear combination of common physical words whose evaluation on block j gives X_j for every j.

The source first invokes the quantum Wielandt theorem: after at most D^4 blockings, each normal tensor becomes injective. It then cites David2006 for the cross-block step: if L is a blocking length after which every normal block is injective, then after $3(L+1)(D-1)$ blockings the tensor is block injective. The proposition records the resulting bound in the following form: after blocking at most $3D^5$ spins, every tensor in canonical form is in block-injective canonical form. This is the assertion labelled Cirac–Perez-Garcia–Schuch–Verstraete 2017, Proposition IV.3. The local source is `Papers/1`

[1] For traceability, this note corresponds to issue #1043. Related formal work is recorded in issue #587 and issue #822; earlier proof attempts are PR #682 and PR #813.

606.00608/MPDO-22-12-17-2.tex, lines 271–345. The cited reference is the paper's item David2006, namely Pérez-García–Verstraete–Wolf–Cirac [PGVWC07].

2 The formal replacement

The formal treatment does not yet derive this proposition from the canonical-form and basis-of-normal-tensors hypotheses. Instead, it makes the finite-length conclusion explicit. The horizontal canonical-form data record the following trace-separation assumption: there is a length N such that, if matrices $\Delta_j \in M_{D_j}(\mathbb{C})$ satisfy

$$\sum_{j=1}^{g} \operatorname{tr}(\Delta_j A_j^w) = 0 \qquad \text{for every word } w \text{ of length } N,$$

then $\Delta_j = 0$ for every j. By nondegeneracy of the trace pairing on the product algebra, this is the dual form of the finite-length spanning property above.

Several exact sufficient hypotheses have been proved. The first is the finite-length product-algebra span itself: the tuples

$$w \longmapsto (A_1^w, \dots, A_g^w)$$

span $\bigoplus_j M_{D_j}(\mathbb{C})$ for one common length. The second is an entrywise linear-independence criterion: the scalar functions $w \mapsto (A_j^w)_{\alpha\beta}$, indexed by the block and by a matrix entry in that block, are linearly independent at one common length. This implies the product-algebra span, hence the trace-separation property. The third is a selector-word formulation of Proposition IV.3: after a common blocking length each block is injective, and after a further finite length there are word polynomials which are the identity on one chosen block and vanish on all the others. Concatenating an injective prefix with such a selector suffix gives the full product-algebra span.

In the formal files, these three finite-length criteria suffice to produce the horizontal canonical-form trace-separation conclusion.[2] Thus Proposition

[2] The criteria are named WordTupleSpanTop, wordEntryFamily, and PropBlockInjective. The trace-separation assumption is HorizontalCFData.biCF. The constructors from these hypotheses to horizontal canonical-form data are in TNLean/MPS/MPDO/BiCFDerivation.lean. Relevant implications include hasBiCF_of_wordTupleSpanTop, hasBiCF_of_wordEntryFamily_linearIndependent, hasBiCF_of_propBlockInjective, and corresponding HorizontalCFData constructors such as horizontalCFData_of_propBlockInjective. The blueprint records the same state in blueprint/src/chapter/ch02b_mpdo.tex, lines 708–787.

IV.3 has not been asserted without proof. The formal statements prove that several finite-length witnesses are sufficient, while the derivation of such a witness from the full canonical-form and basis-of-normal-tensors hypotheses remains open.

3 The counterexample boundary

A tempting shortcut is false. One cannot recover the finite-length trace-separation witness from blockwise injectivity, left-canonicality, and nonzero weights alone. Take two one-dimensional scalar blocks over a one-letter alphabet, with both block tensors equal to 1 and with weights 1 and 2. Each block is injective and left-canonical, and the weights are nonzero and distinct. However every word evaluation is the same on the two blocks. The tuple span is the diagonal line in $\mathbb{C} \oplus \mathbb{C}$, not the whole product algebra. Equivalently, choosing $\Delta_1 = 1$ and $\Delta_2 = -1$ makes the trace pairing vanish on every word while the tuple (Δ_1, Δ_2) is nonzero.

This example is formalized in `TNLean/MPS/MPDO/BiCFDerivation.lean`, lines 1363–1463, and is recorded in `docs/counterexamples.md`, lines 29–35. It shows that the additional finite-length witness in the formal hypotheses is genuinely stronger than the other horizontal canonical-form assumptions. It does not refute Cirac–Perez-Garcia–Schuch–Verstraete 2017 Proposition IV.3. The duplicate scalar blocks are related by similarity and phase, and therefore do not form a minimal basis of normal tensors in the sense of the source characterization.

4 Conclusion

The conclusion is provisional and should not be read as a source erratum. The formal statements replace the paper's finite-length block-separation theorem by explicit finite-length span, linear-independence, or selector hypotheses. A small counterexample shows that those hypotheses cannot be derived from the weaker assumptions alone. The full source theorem remains plausible under the actual canonical-form and minimal basis-of-normal-tensors hypotheses.

What remains to be formalized is the David2006/Cirac–Perez-Garcia–Schuch–Verstraete 2017 finite-length block-separation argument: from minimality of the basis-of-normal-tensors blocks, together with single-block injectivity after Wielandt blocking, one must produce a common finite selector or entrywise linear-independence witness, with the paper's bound

$3(L+1)(D-1)$ and hence the stated $3D^5$ corollary. Until that argument is available, the finite-length witness should remain an explicit hypothesis.

References

[CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: Annals of Physics 378 (2017), 100–149, 2016.

[CPGSV17] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. *Annals of Physics*, 378:100–149, 2017.

[PGVWC07] David Pérez-García, Frank Verstraete, Michael M. Wolf, and J. Ignacio Cirac. Matrix product state representations. *Quantum Information and Computation*, 7(5–6):401–430, 2007.